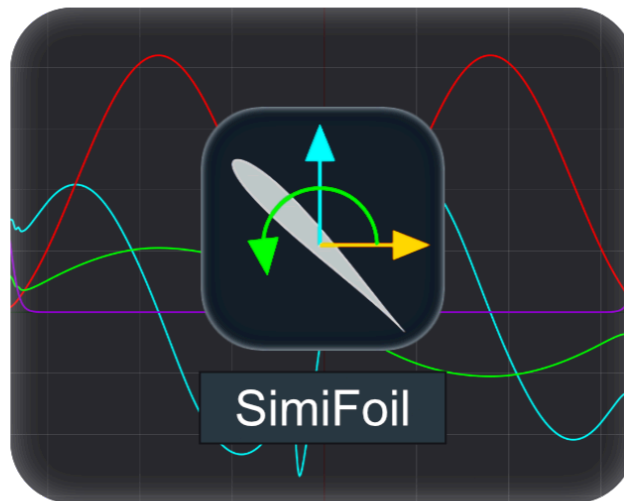




Finally: A Real-Time Aerodynamics Tool that Handles Deep Stall and Reverse Flow



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Most aerodynamic tools stop working after $\sim 20^\circ$ angle of attack. But for flight simulators and game development, we need to understand what happens during spins, tailslides, and aerobatics.

I originally began developing **SimiFoil** while working on my flight simulator project **SimiFlight**. What started as an internal tool quickly evolved into a standalone, lightweight aerodynamic curve generator for flight simulators, RC aircraft, and game engines.

👉 Try the free Web Demo: [SimiFoil](#)

SimiFoil generates **plausible, continuous aerodynamic coefficients across the full 360° angle-of-attack range**, including post-stall and reverse-flow regimes. The goal is not CFD-level accuracy, but **stable, simulation-friendly data** suitable for real-time applications and Blade Element Theory (BET).

While SimiFoil handles **classical and ideal airfoil shapes** well, it is also designed to handle **non-standard and unconventional profiles**, such as:

- Flat plates with thickness or rounded edges
- Airfoils with blunt or thick trailing edges
- Highly modified or non-classical geometries

While the results are not perfect for such profiles, the generated curves remain **physically plausible and numerically stable**, which is critical for simulation and control systems.

To validate the model, I compared the generated curves against experimental wind tunnel data, including the **NACA Technical Note 3361 (1955, Critzos et al.)**, which investigates the NACA 0012 airfoil over the full 0° – 180° (and even 360°) angle-of-attack range. The agreement is remarkably close, especially considering the real-time constraints and analytical nature of the model.

The Web Demo runs directly in the browser and allows you to:

- Analyze lift, drag, and moment coefficients
- Explore polar plots (C_l vs. C_d)
- Design airfoils and experiment in a visual wind tunnel

File export is intentionally disabled in the web demo.

A standalone desktop version with **CSV, LUT, and Xfoil/XFLR5 export** is already **implemented** and is currently being prepared for release.

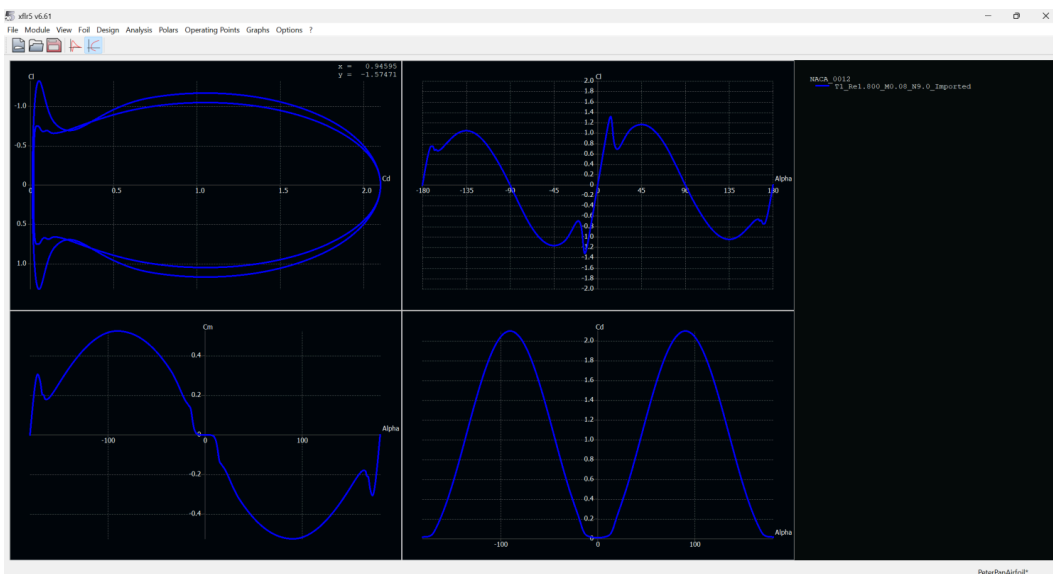
Development is ongoing. I plan to continue refining the aerodynamic model and expanding the feature set based on feedback and real-world use cases.

SimiFoil: Real-time visualization and XFLR5 Xfoil import.



Real-time visualization and curve generation inside SimiFoil

The SimiFoil Dashboard: Analyzing a NACA 0012 profile. The left panel handles configuration, the center shows 360° Lift (Cl), Drag (Cd), and Moment (Cm) curves, and the right panel visualizes real-time forces in a virtual wind tunnel.



Example of aerodynamic curves generated with SimiFoil and exported to XFoil format, shown here inside XFLR5.

The screenshot shows aerodynamic curves generated with SimiFoil and exported in XFoil-compatible format, visualized in XFLR5.

The goal is not CFD-level accuracy but to provide **stable, continuous aerodynamic data across the full 360° angle-of-attack range**, including post-stall behavior, suitable for real-time simulation and design iteration.

This makes it possible to use the generated data directly in existing analysis tools and simulation workflows.

If you work in flight simulation: **How do you currently handle post-stall and reverse-flow aerodynamics?** I'd love to hear your thoughts.

#FlightSimulation #Aerodynamics #Physics #GodotEngine #GameDev #Engineering
#SimiFoil #XFoil #XFLR5